NAME: SRUTHI RITHIKA. P
REGNO: 192521407
COURSE: operating system
COURSE CODE: CSA 0401

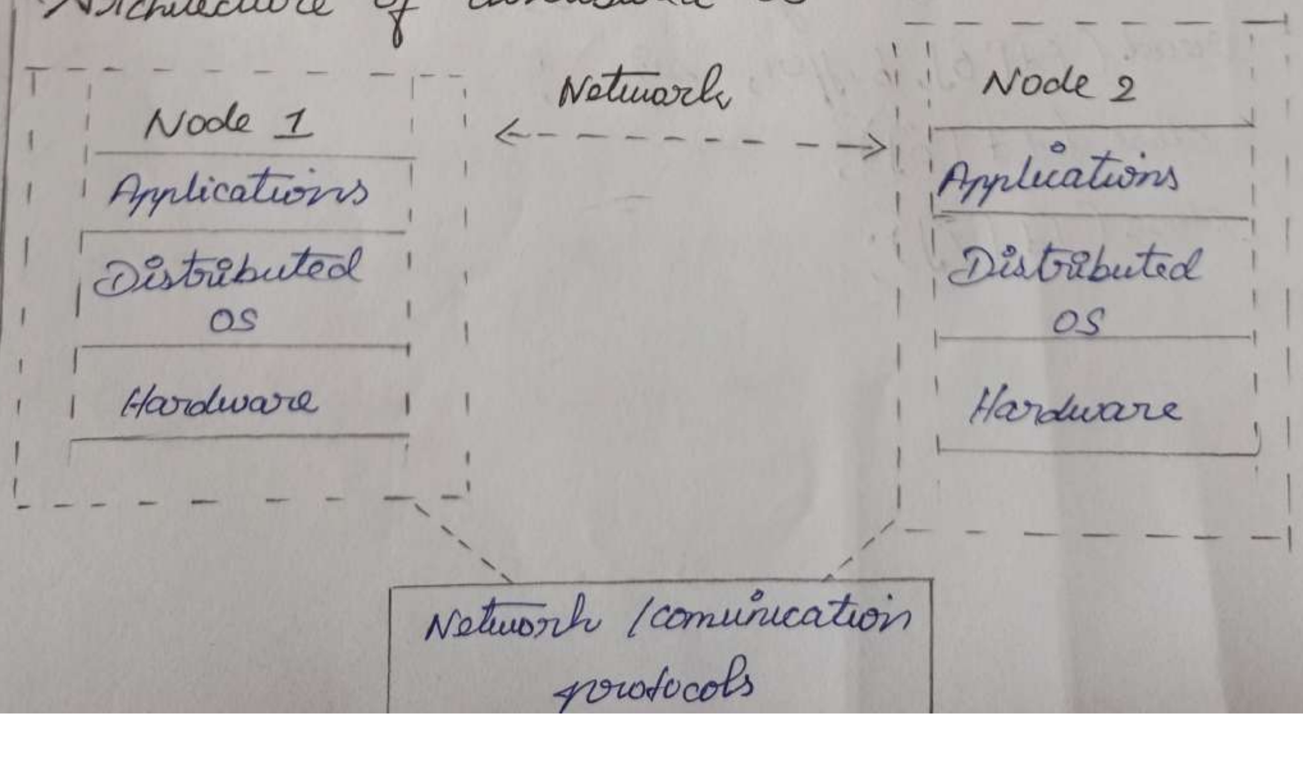
Short Answers

4)

a) Classification and comparison

Distributed OS	Centralized OS
* Manges multiple interconnected computer /nodes.	Manges a single central computer / system.
* Resource are distributed among different nodes.	* Resources are located mainly in one central system.
* Provides transparency users may feel they are using one system.	* users directly access the central system.
* Failure of one node may not stop the entire system.	* Failure of central system can affects the whole system.

b) Architecture of distributed OS



Node 3
Distributed OS
Hardware

Node 4
Distributed OS
Hardware.

c) sequence of system calls in IPC.

pipe ()
↓
fork ()
↓
close () → close unused pipe end
↓
write () → sender writes data
↓
read () → Receiver reads data
↓
close ()

Example:

```
pipe(fd);
fork ();
write (fd [1], message, size);
read (fd[0], buffer, size);
close (fd [0]);
close (fd [1]);
```

5)

a) Major Functions of an operating system

1. Process Management /scheduling

* It creates and terminates process.

* schedules process on the CPU.

* Performes context switching.

* Allocates CPU time using algorithms such as FCFS, SJF, priority, and Round Robin.

2. Memory Management.

* Allocates & deallocates memory for process.

* keeps track of used and free memory.

* Provides virtual memory.

* Performes paging and segmentation when required.

3. Device Management.

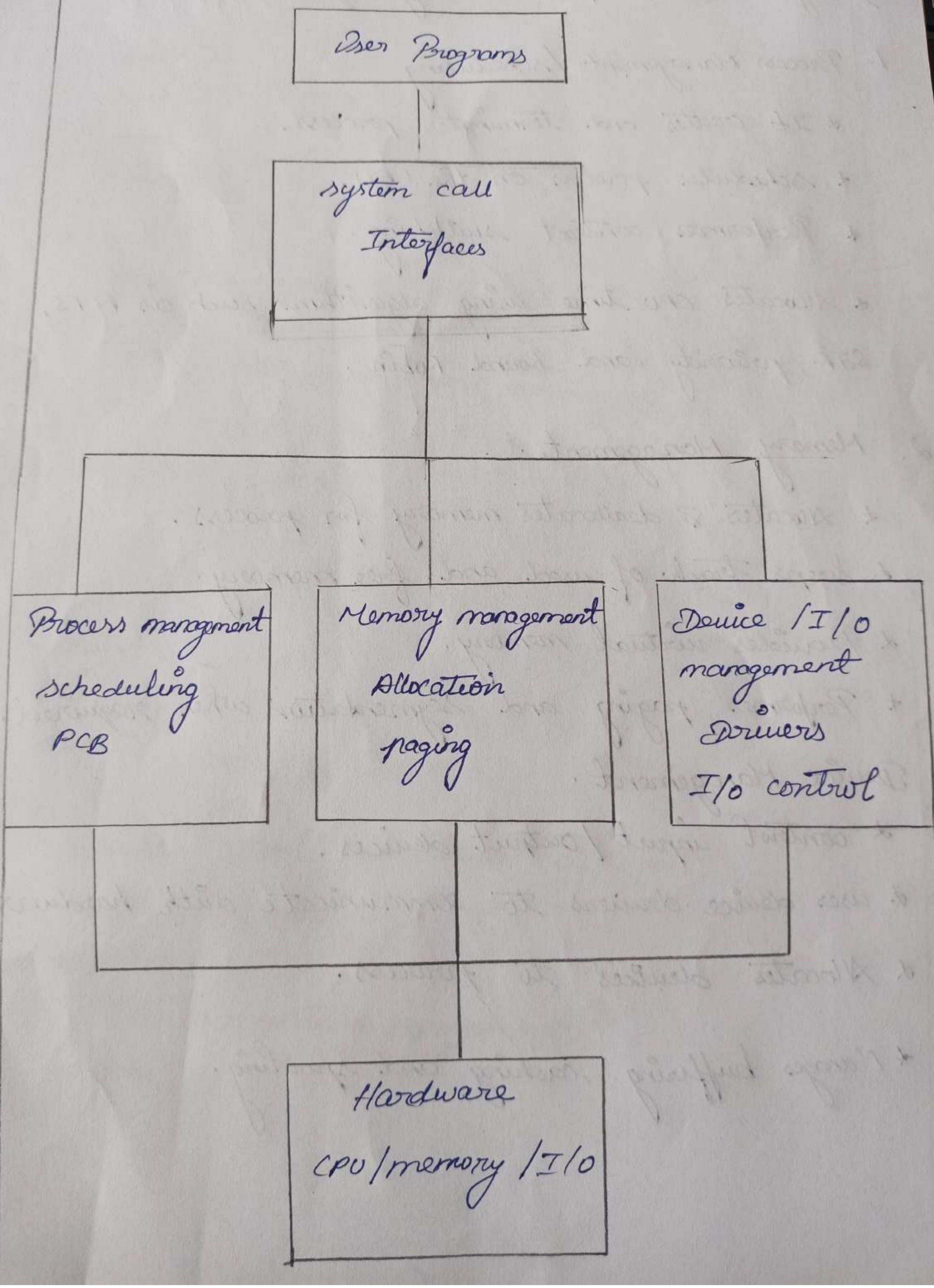
* control input / output devices.

* uses device drivers to communicate with hardware

* Allocates devices to process.

* Manages buffering, caching and spooling.

b) Block Diagram of OS components.



c) system call Examples.

Memory Allocation.

```
brk();
sbrk();
mmap();
munmap();
```

Exa:

```
void *ptr = mmap(NULL, 4096,
            PROT_READ | PROT_WRITE,
            MAP_PRIVATE | MAP_ANONYMOUS,
            -1, 0);
```

Device Management.

```
open();
read();
write();
ioctl();
close();
```

Exa:

```
int fd = open("/dev/device", O_RDWR);
read(fd, buffer, size);
write(fd, buffer, size);
ioctl(fd, command, arugement);
close(fd);
```